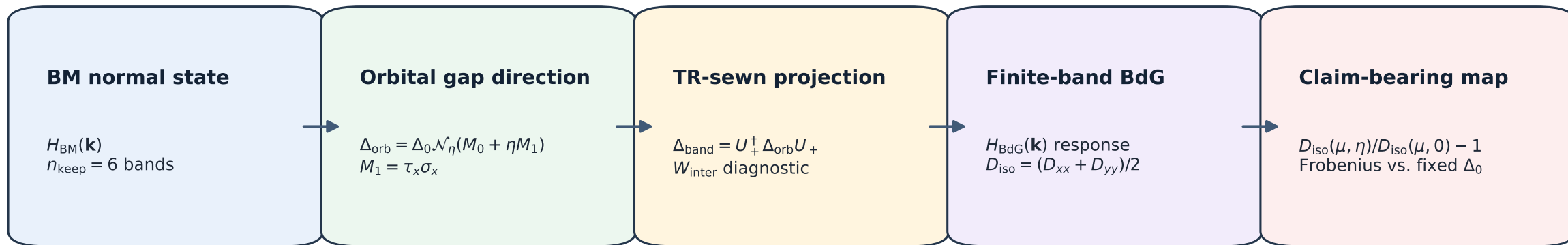


Executable route from orbital pairing structure to a normalized MATBG response signature



Self-audit boundary: normalized ratios carry the mechanism claim; raw eV $\text{\AA}^2$ values and filling proxies remain benchmark controls.